Student Academic Performance Prediction

Project Overview

This project implements an end-to-end Machine Learning pipeline to predict student academic performance based on socio-economic and demographic factors. By analyzing features such as study time, parental education, and sleep duration, the model classifies students into performance tiers (Low, Medium, High).

The primary goal is to identify at-risk students and provide actionable insights for educational interventions.

Technical Stack

- **Languages:** Python 3.10+
- **Libraries:** * **Pandas & NumPy:** Data manipulation and numerical computation.
    - **Scikit-Learn:** Model training, preprocessing, and evaluation.
    - **Imbalanced-Learn (SMOTE):** Handling class imbalance.
    - **Matplotlib & Seaborn:** Data visualization.

The ML Pipeline

The project follows a rigorous data science workflow:

1. Exploratory Data Analysis (EDA)

- Analyzed distribution of numerical features (Math, Reading, and Writing scores).
- Identified correlations between study time and final academic outcomes.

2. Data Preprocessing and Engineering

- **Missing Value Imputation:** Utilized SimpleImputer with a median strategy to handle null values in academic scores and categorical data.
- **Categorical Encoding:** Applied LabelEncoder for features like Race/Ethnicity and Parental Education.
- **Outlier Removal:** Implemented **Interquartile Range (IQR)** filtering to remove noise from numerical data.
- **Feature Synthesis:** Created a composite avg_score feature to represent holistic academic performance.

- **Scaling:** Used StandardScaler to normalize feature distributions for the classification model.

3. Handling Class Imbalance

To ensure the model does not favor the majority class (Medium performers), **SMOTE (Synthetic Minority Over-sampling Technique)** was applied to the training set to balance the Low and High performance categories.

4. Model Selection: Random Forest Classifier

The model utilizes a **Random Forest Ensemble** with the following parameters:

- n_estimators: 200
- max_depth: 10
- random_state: 42

Evaluation Metrics

The model was evaluated using both cross-validation and standard classification metrics:

- **Test Accuracy:** Approximately 68.7%
- **Cross-Validation Score:** Approximately 70.4%
- **Key Insight:** High Recall for the "Medium" class (approximately 96%), indicating reliable identification of the core student group.

Visualizations

The notebook includes comprehensive plots:

- **Confusion Matrix:** To visualize misclassifications between performance tiers.
- **Tree Visualization:** A limited-depth plot of a decision tree within the Random Forest to explain model logic.

How to Run

1. **Clone the repository:**

Bash

```
git clone https://github.com/your-username/student-performance-prediction.git
```

cd student-performance-prediction

2. **Install dependencies:**

Bash

pip install pandas numpy scikit-learn matplotlib seaborn imblearn

3. **Execute the notebook:** Open STUDENT_SCORE_PREDICTION.ipynb in Jupyter Notebook or VS Code and run all cells.

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